

Avant Actuation



Spokane Valley Tech
Peyton Ward & Sarah Orebaugh
2025 - 2026

Table of Contents

1. Executive Summary.....	2
2. Company Profile.....	3
Mission Statement.....	3
Governance.....	3
Location.....	3
Goals.....	3
3. Industry Analysis.....	4
Trends and Growth Rates.....	4
Strategic Opportunities.....	4
4. Target Market.....	5
Size and Potential.....	5
Accessibility.....	5
5. Competitive Analysis.....	6
Key Rivals.....	6
Our Competitive Advantage.....	7
6. Marketing and Sales.....	7
Marketing Execution.....	8
Direct Sales Strategy.....	8
Public Relations.....	9
7. Operations.....	9
Day-to-Day Processes.....	9
Technology and Facilities.....	10
8. Management and Organization.....	10
Key Personnel.....	10
Board of Advisors.....	11
9. Long-Term Development - Year 3-5 Vision.....	11
Market Expansion.....	11
Product Intellectual Property and Transparency.....	11
Operational Scaling.....	11
Exit Strategy: Sustainability & Ownership.....	12
10. Financials.....	13
Year 1 Monthly Cash Flow Projection.....	14
Year 1 Monthly Income Statement (Summary).....	14
Yearly Income Statements.....	15
11. Appendix.....	15
Exhibit A: Letter of Intent (LOI).....	15
Exhibit B: Technical Design and Prototyping.....	16
Exhibit C: Equipment and Facility Requirements.....	16
Major Industry Sources.....	16

Prosthetics Supply Business Plan

1. Executive Summary

The Opportunity: The prosthetic device market is astonishingly archaic, driven by extreme pricing and vertical integration that prioritizes corporate profit over patient mobility. Current industry leaders have created a vertical integration oligopoly, forcing practitioners into expensive, proprietary contracts for parts that have not seen innovation in years, and times, in decades. With the global market growing at a 5.5% CAGR, there is a massive, unanticipated demand for a provider that uses open-source compatibility and modern fabrication to break these oligopolies. Avant Actuation is positioned to bridge this gap, offering high-performance solutions at a price point that was previously considered impossible in the current market.

The Solution: Avant Actuation utilizes advanced manufacturing techniques—including 3D prototyping via Fusion 360 and inventory management through Katana—to drastically reduce overhead and production costs. By focusing on field-repairability and universal compatibility, we provide practitioners with a reliable, cost-effective alternative to industry giants without sacrificing patient needs.

Target Market Highlights: We focus on the business-to-business (B2B) practitioner market, navigating FDA requirements to sell directly to clinics. While the global industry is nearly \$7 Billion, our initial focus is the Pacific Northwest regional market, where we aim to secure our first 5–10 pilot clients within Year 1 to establish a proof of concept for our manufacturing model.

Financial Potential: Avant Actuation is projected to achieve a break-even point by Month 12, with a Year 1 revenue goal of \$40,000 during the prototyping and clinical testing phase. By Year 5, as we scale operations to larger facilities and expand into national distribution, we project total revenues of \$850,000 with a target net profit margin of 20–30%.

The Team: Led by Sarah Orebaugh (CEO) and Peyton Ward (COO), our leadership combines strong engineering and fabrication skills with technical logistics and customer service expertise. Supported by a robust Board of Advisors specializing in biocompatibility

and prosthetic fabrication, the team is uniquely positioned to bridge the gap between complex engineering and clinical application.

2. Company Profile

Mission Statement

The prosthetic device market is currently characterized by prohibitive pricing structures that appear disproportionate to actual manufacturing costs. This trend of arbitrary price escalation by medical device manufacturers creates significant financial barriers, limiting patient access to essential mobility technology. This is proving to be an opportunity to serve the amputee community by solving these decades-old grievances, via new-age manufacturing and open market business operations.

Avant Actuation was started on September 24th, 2025, by Sarah Orebaugh and Peyton Ward

Governance

Business Structure	Ownership Percentage	Key Stakeholders
Limited Liability Company	51% CEO 49% COO	Sarah Orebaugh

Location

The primary business operations will be located in  Spokane .

Goals

Type	Short-Term (Year 1)	Mid-Term (Years 2-3)	Long-Term (Years 4-5)
Market	Secure first 5–10 pilot clients and establish a local brand presence.	Expand the service area to include the greater regional market.	Become the primary regional provider and authority in the sector.

Type	Short-Term (Year 1)	Mid-Term (Years 2-3)	Long-Term (Years 4-5)
Product	Finalize supplier contracts and establish a reliable fulfillment process.	Launch a proprietary product line of high-end active joints and supporting parts	Establish an R&D partnership to innovate new industry solutions.
Financial	Achieve a break-even point by the end of Month 12.	Sustain a 15% year-over-year revenue growth.	Achieve and maintain a 20-30% net profit margin.

3. Industry Analysis

Trends and Growth Rates

- Current Industry Status: Approximately \$9.65 Billion Industry
- Key Trends: Shifts to microprocessor joints, AI-driven sensors, and custom-made 3D printed parts. A yearly growth of 5.5% via Compound Annual Growth Rate (CAGR)
- Growth Projections: The prosthetics industry is expected to grow at a CAGR rate of 5.5% and grow to approximately \$12.61 Billion by 2030

Strategic Opportunities

The prosthetics market is on the cusp of a "two-front" innovation strategy, which Avant Actuation is uniquely positioned to execute. First, we are replacing archaic, labor-intensive manufacturing methods with high-precision 3D scanning and additive manufacturing, drastically reducing lead times and production waste. Second, we are leveraging a "Market-Open" business model. By purposely avoiding the vertical integration and restrictive contracts used by industry giants. This dual approach allows us to provide practitioners with the flexibility they have been denied for decades, creating a massive opportunity for rapid market entry.

4. Target Market

Size and Potential

- **Primary Demographic:** Our target demographic is Certified Prosthetist-Orthotists (CPOs) and clinical practitioners. Due to FDA regulations, prosthetic components must be sold directly to licensed practitioners who handle the final assembly and fitting, ensuring patient safety and proper device alignment
- **Total Available Market (TAM):** The global prosthetics and orthotics market is valued at approximately \$9.65 Billion in 2026
- **Serviceable Available Market (SAM):** Our immediate focus is the North American market, which holds a dominant 40% share of the global industry (approx. \$3.86 Billion). Specifically, we are targeting the Pacific Northwest regional clinic network to establish our initial proof of concept

Accessibility

We intend to sell to practitioners. By doing so, our business will be following the laws set by the FDA. Other benefits of selling our product directly to practitioners.

- **Quicker Feedback**
- **Better Partnerships**
- **Higher Trust with Clinic and Consumer**

5. Competitive Analysis

Key Rivals

Competitor	Strengths	Weaknesses	Your Strategy to Compete
Fillauer	· Strong heritage in US manufacturing · Diverse product catalog (feet, knees, orthotics) · Trusted by traditional clinicians	· Slower to innovate compared to tech giants · Traditional manufacturing methods keep costs high · Marketing is often outdated	· Utilize modern manufacturing (e.g., 3D printing) to lower overhead costs · Market directly to younger, active demographics who value durability over tradition
Ottobock	· Global market leader with high brand recognition · Cutting-edge technology (e.g., C-Leg, bionics) · Massive R&D budget	· Extremely high price point (inaccessible to many) · Complex reimbursement/insurance processes · Repairs can be slow and proprietary	· Focus on the "gap" market: users who need reliability, but cannot afford premium bionics · Emphasize "field repairability" and low-cost maintenance
Össur	· Innovation leader in non-invasive orthopedics. · Strong "Life Without Limitations" branding · High-quality silicone liners and sockets	· Premium pricing structures · Heavy focus on high-activity/athlete segments, sometimes neglecting average users · Proprietary systems limit compatibility	· Offer universal compatibility with other standard parts · Price your product aggressively to capture the mid-tier market that Össur ignores

Competitor	Strengths	Weaknesses	Your Strategy to Compete
Hanger Clinic	· Largest O&P (Orthotic and Prosthetic) provider in the US · Massive network of 800+ patient care clinics · Strong referral relationships with hospitals	· Primarily a service provider, not a product manufacturer · High markup on devices sold to patients · Patient experience can feel "corporate" and impersonal	· If B2B: Position your product as a cost-saving alternative for their inventory · If B2C: Bypass the clinic markup by selling direct-to-consumer (where legal) or partnering with independent clinics

Our Competitive Advantage

Avant Actuation combines new age manufacturing with agile inventory software and an understanding of the goal of the prosthetics industry to lower production costs for the good of the patient. Unlike our legacy competitors, who use vertical integration to force sales, our company strives to create universally compatible, reliable, field-repairable, high-quality parts at a significantly lower price, naturally competing with legacy products without forcing sales.

6. Marketing and Sales

Strategic Approach

Positioning: Avant Actuation is positioned as the "Practitioner's Partner." While legacy corporations use vertical integration to lock clinics into expensive, proprietary ecosystems, we offer "Clinical Freedom" through universal compatibility and field-repairable components. We are the most cost-effective and reliable alternative for modern, high-activity patients.

Pricing Strategy: By leveraging 3D prototyping and agile manufacturing, we will price our components 25–40% below the current market average. This allows clinics to increase their own margins while providing patients with premium-grade technology that was previously financially inaccessible.

Marketing Execution

Marketing Channel	Activity	Budget Allocation	Key Metric
Direct Sales (B2B)	"Lunch & Learn" demos at PNW clinics; Booth presence at AAOP and AOPA regional meetings.	\$6,000	Number of Pilot Partnerships (Target: 10 in Year 1)
Digital Presence	SEO-optimized technical portal for CPOs (3D CAD previews, compatibility charts); LinkedIn B2B ads.	\$2,500	Cost Per Lead (CPL) & Spec-Sheet Downloads
Public Relations	Local media features (KXLY/KCPQ) focus on "Spokane-made innovation" and patient success stories.	\$1,000	Media Impressions & Brand Sentiment
Strategic Referral	Partnership with patient advocacy groups (e.g., Amputee Coalition) to drive patient-side demand.	\$500	Referral Inquiries from Practitioners

Direct Sales Strategy

Our primary sales tactic is the "Clinical Pilot Program." We will provide a 30-day "No-Risk Trial" of our joints to 10 select clinics in the Spokane and Seattle areas. This bypasses the CPO's fear of switching brands by letting them stress-test our field-repairability in a real-world clinical setting.

Public Relations

By partnering with local news stations like KXLY and KCPQ, we will focus on the human element: a Spokane-based company reducing the financial burden on local amputees. This creates a "community pull" where patients ask their doctors for Avant Actuation products by name.

7. Operations

Day-to-Day Processes

This is information on how Avant Actuation will operate Day-to-Day

- **Procurement:** We will use Katana to track stock levels and set automatic reorder alerts. When parts run low, we will research suppliers to find the best price and quality. We will create Purchase Orders in Katana and email them to our suppliers. Katana will automatically sync these financial details to QuickBooks to keep our accounting accurate. Once parts arrive at the plant, we will verify the count, mark the order as received in Katana to update our stock numbers, and pay the supplier's bill through QuickBooks.
- **Inventory Management:** We would manage our inventory using the Katana software, which is the best fit for live inventory data, and use inside of a manufacturing setting. When we receive an order, we would request the piece from our inventory building. We would hold our inventory inside the Old Kaiser Aluminium Plant in North Spokane*. This place has a large square footage for not only storage, but it can also serve as our manufacturing plant (Years 5-10)*
- **Fulfillment and Logistics:** We would ship using the healthcare division of UPS, they are fast and can ship throughout the US

Errors: If a hiccup were to occur, like if a prosthetic arrived at a practitioner's office damaged/broken, we could replace it relatively quickly

Technology and Facilities

- Software: Fusion 360, Katana, QuickBooks
- Equipment: Shelving, Forklifts, Boxes, Label Printer, Computers, Internet Infrastructure
- Facilities: 📍 1711 E Trent Ave
 - Size: 2880 Squarefeet of shop space, 1420 Squarefeet of office space.
 - Purpose:
 - Prototyping
 - Office Space
 - Management

8. Management and Organization

Key Personnel

Name	Title/Role	Core Responsibilities	Compensation (Salary/Equity)
Sarah Orebaugh	CEO/General Manager/Marketing	● Ensures Plan is Met ● Setting Long-Term Goals	Year 1: \$0 (Deferred) / Year 2: \$30,000 + 51% Equity
Peyton Ward	COO/Sales Director	● Logistics ● Customer Service ● Marketing Design	Year 1: \$0 (Deferred) / Year 2: \$30,000 + 49% Equity

Board of Advisors

Our board of advisors meets once every two weeks alongside our CEO, Sarah Orebaugh, and COO Peyton Ward. This board meets to assess the business's current status and advise leadership on next steps.

- Arlana Nielsen, Expertise: Product Bio Compatibility with the human body
- Ambrose Cavegn, Expertise: Prosthetic fabrication and upkeep
- Chad Orebaugh Expertise: Product Research, Development, and Prototyping
- Alvaro Cuavas Expertise: Product manufacturing and estimation

9. Long-Term Development - Year 3-5 Vision

Market Expansion

By Year 5, the company will initiate international distribution, targeting strategic market entry into Canada and Mexico to broaden our North American footprint.

Product Intellectual Property and Transparency

We will implement a robust IP strategy by securing patents for proprietary internal components. This approach protects the brand from industry incumbents while maintaining a policy of technical transparency. By allowing stakeholders to understand our product mechanics, we aim to strengthen trust and facilitate deeper engagement with both clinical partners and the end consumer.

Operational Scaling

To support increased demand and optimize manufacturing lead times, we will transition operations to a larger-scale facility, such as the Hawthorne Road industrial complex. This expansion will provide the necessary infrastructure to scale production volume and enhance overall operational efficiency.

Exit Strategy: Sustainability & Ownership

Avant Actuation is committed to remaining a privately held, independent entity. Unlike competitors who are beholden to external shareholders and quarterly profit demands, our private structure allows us to prioritize patient outcomes and long-term innovation. We do not intend to seek venture capital or private equity, ensuring that 100% of the decision-making power remains with the founders.

Because we rely on loans and grants rather than equity investors, our "Exit" is defined by reaching financial self-sufficiency and the retirement of all debt obligations.

Debt Retirement Plan: Our primary financial goal is the aggressive repayment of the SBA 7(a) Express Loan and STCU Loan using operational cash flow. By retiring these debts by Year 5, we eliminate interest expenses and maximize the company's net valuation.

Grant Utilization & Compliance: We will leverage the NIH/NSF SBIR and Evergreen Manufacturing grants to fund high-risk R&D and scaling. Because these are non-dilutive, they allow us to innovate without surrendering any ownership stake.

Founder Reinvestment: Instead of sharing profits with external stakeholders, all surplus revenue will be split between two primary channels:

- **Internal R&D:** Funding the next generation of 3D-printed prosthetic components (as detailed in Exhibit B).
- **Operational Reserve:** Building a "War Chest" to fund our Year 5 transition to the Hawthorne Road industrial complex without requiring further significant debt.

Long-Term Hold: We are building a "Legacy Brand." Our success is measured by our ability to remain an independent regional authority in the Pacific Northwest, maintaining absolute control over our manufacturing quality and pricing structures.

10. Financials

The financial projections below are conservative and based on an honest assessment of the market. Avant Actuation will utilize the accrual basis of accounting(using QuickBooks) to ensure accurate tracking of inventory and accounts payable/receivable with our clinic partners

Source	Type	Amount	Purpose	Timing
SNAP Financial	Microloan	\$25,000	Initial prototyping costs.	Q1
SBA 7(a) Express	Loan	\$75,000	FDA fees and consulting.	Q2
NIH / NSF SBIR Phase I	Grant	\$275,000	Clinical and safety testing.	Q3
STCU	Loan	\$50,000	Ongoing inventory purchases.	Q1 2027
Evergreen Manufacturing	Grant	\$250,000	Washington-based manufacturing scaling.	Year 3 (2028)

Year 1 Monthly Cash Flow Projection

Month	Starting Cash	Cash Inflows (Sales)	Cash Outflows (OpEx, Inventory)	Net Cash Flow
Month 1	\$75,000	\$0	\$15,000	-\$15,000
Month 3	\$48,000	\$0	\$12,000	-\$12,000
Month 6	\$2,500	\$1,000	\$1,500	-\$10,500
Month 9	\$331,400	\$4,500	\$15,300	-\$10,800
Month 12	\$300,000	\$12,000	\$15,000	-\$3,000

Year 1 Monthly Income Statement (Summary)

Month	Revenue	Cost of Goods Sold	Gross Profit	Operating Expenses	Net Income
Month 1	\$0	\$0	\$0	\$15,000	-\$15,000
Month 3	\$0	\$0	\$0	\$12,000	-\$12,000
Month 6	\$1,500	\$600	\$900	\$12,000	-\$11,100
Month 9	\$4,500	\$1,800	\$2,700	\$13,500	-\$10,800

Month 12	\$8,000	\$3,200	\$4,800	\$15,000	-\$10,200
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Yearly Income Statements

Category	Year 1	Year 3	Year 5
Total Revenue	\$40,000	\$350,000	\$850,000
Cost of Goods Sold	\$16,000	\$140,000	\$340,000
Gross Profit	\$24,000	\$210,000	\$510,000
Operating Expenses	\$150,000	\$120,000	\$200,000
Depreciation	\$5,000	\$15,000	\$30,000
Interest Expense	\$11,500	\$10,000	\$5,000
Earnings Before Taxes (EBT)	-\$142,500	\$65,000	\$275,000
Taxes (Rate)	\$0	\$13,000	\$57,750
Net Income	-\$142,500	\$51,350	\$2

II. Appendix

Exhibit B: Technical Design and Prototyping

Software Platform: Autodesk Fusion 360

Current Status: Alpha Prototyping Phase

The following renderings represent the current revision of the knee.

Exhibit C: Equipment and Facility Requirements

To achieve the production goals outlined in Section 7, the following equipment has been sourced for purchase upon receipt of the SBA 7(a) Express Loan.



Item	Purpose	Estimated Cost
Industrial 3D Printer	High-strength carbon fiber nylon printing	\$5,500
Katana MRP License	Inventory and production tracking	\$1,200/yr
Post-Processing Station	Finishing and smoothing components	\$2,100
Quality Testing Rig	Stress-testing joints for safety compliance	\$4,000

Major Industry Sources

Market Size & Growth: Fortune Business Insights. "Limb Prosthetics Market Size, Share Growth Analysis [2034]."

Regulatory & Compliance: U.S. Food and Drug Administration (FDA). "Medical Device User Fee Amendments (MDUFA)."

Manufacturing Innovation: MDPI. "Review of Robotic Prostheses Manufactured with 3D Printing: Advances, Challenges, and Future Perspectives."

Competitor Financial Benchmarks: U.S. Securities and Exchange Commission (SEC). Filings for Hanger, Inc. and PSYONIC, Inc.

Legacy Industry Practices: Federal Trade Commission (FTC). "Federal Trade Commission Orders Otto Bock To Unwind Consummated Merger."

Policy & Market Enclosure: Advanced Medical Technology Association (AdvaMed). "2024 Year in Review."